Calibration of Convex Surrogate Losses via Property Elicitation

Jessie Finocchiaro

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Dr. Ajunwa is an expert on employment and labor law.

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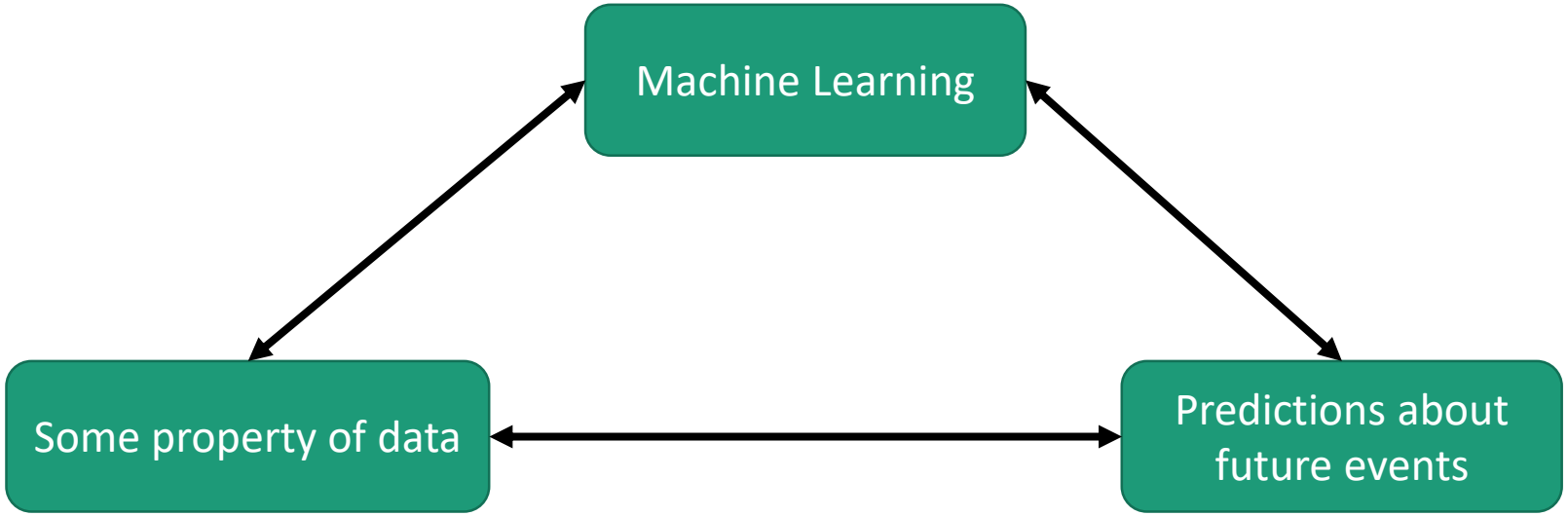
Introduction



Outline

- Background
- Surrogates
  - Calibration
  - Properties
  - Necessary and Sufficient Conditions
- Case Study: Abstain
- Dimensionality
- Conclusion

Empirical Risk Minimization (ERM)

Assumption: data comes i.i.d. from P over $(\mathcal{X}, \mathcal{Y})$.

Goal: minimize risk.

$$Risk(h) = \int L(h(x), y) dP(x, y)$$

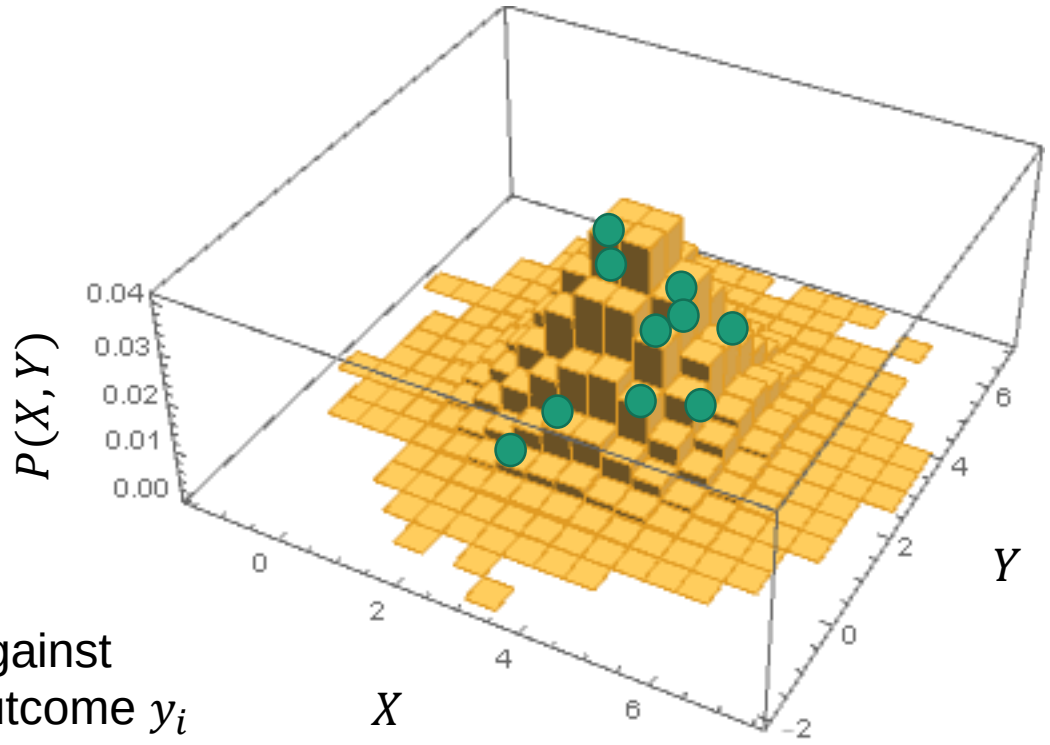
Problem: don't know P.

Settle for minimizing *empirical risk.*

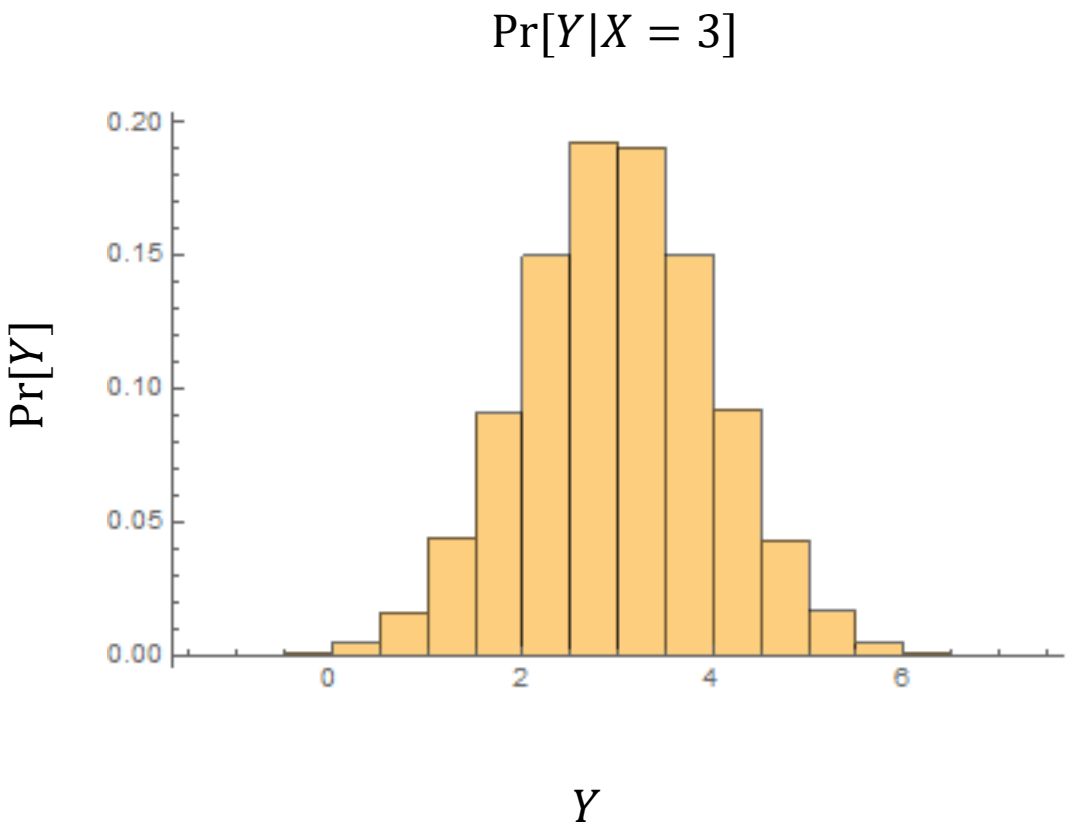
Take m inputs $x_i \in \mathcal{X}, y_i \in \mathcal{Y}$.

$$ERisk(h) = \underbrace{\frac{1}{m}\sum_{i=1}^{m}}_{\text{Average}} \overbrace{L(\underbrace{h(x_i)}_{\text{Prediction given feature } x_i}, y_i)}^{\text{Loss}}$$

Against outcome y_i



Conditional Probability



Form hypothesis $h: \mathcal{X} \to \mathcal{R}$.

Want to learn $h(X)$ minimizing $\mathbb{E}_{(X,Y)\sim P} L(h(X), Y)$.

If $h(x)$ minimizes $L(r, Y)$ over r for all $x \in \mathcal{X}$, obviously true.

Abstract $h(x)$ to r, look at distribution on simplex, Δ_y.

Setting

Classification-like problems

- Multiclass classification (with reject option)
- Ranking
- Top-k classification

Notation

- Finite outcomes $|\mathcal{Y}| \coloneqq n$
- Report set $|\mathcal{R}| \coloneqq k$
- Probability Distribution over $\mathcal{Y}$: $\mathrm{p} \in \Delta_{\mathcal{Y}}$
  - $p_y = \Pr[Y = y]$

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Surrogates

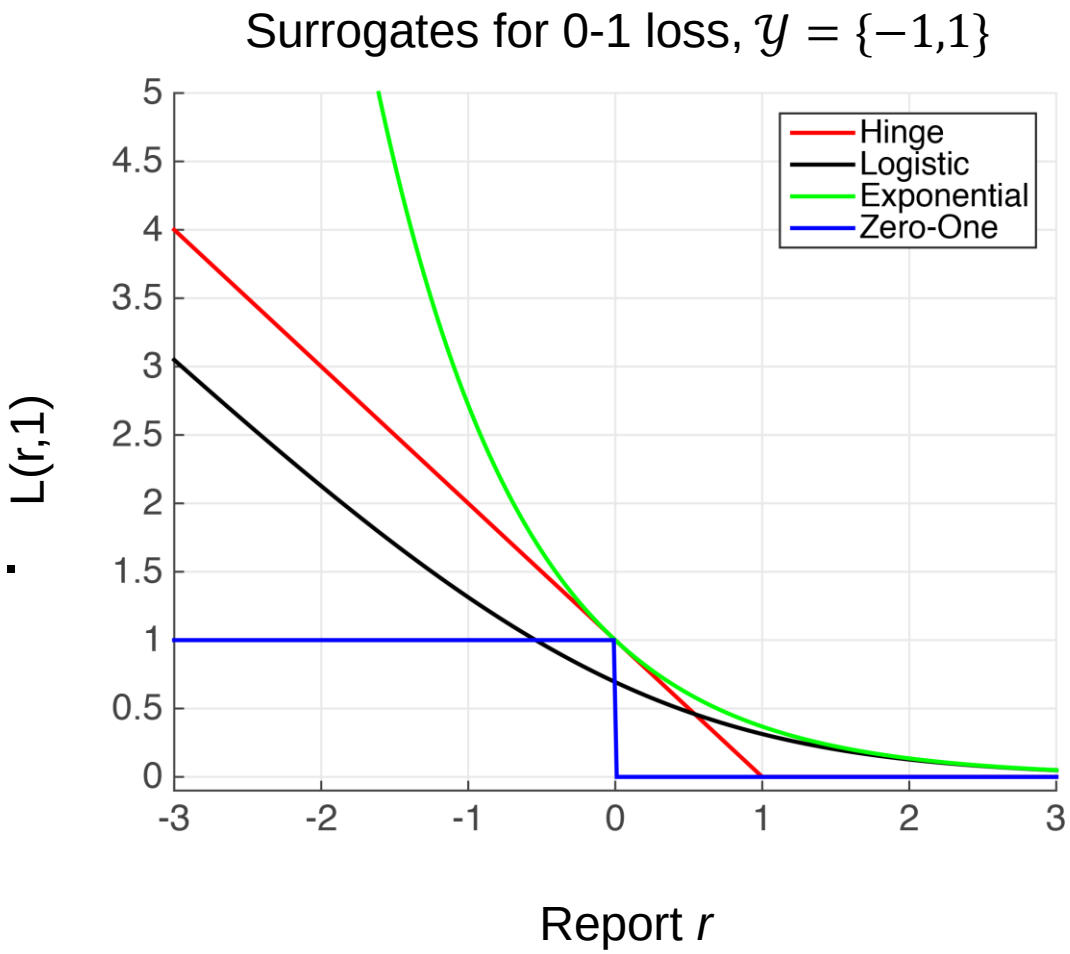
Loss functions measure error.

Created with a task in mind.

- Often discrete.

Discrete losses hard to optimize.

Surrogates should approximate the original loss well.



Calibration

A calibrated loss L is "good approximation" of discrete loss $\ell(r, y)$.

Let ℓ be a discrete loss. A surrogate loss function $L: \mathbb{R}^d \times \mathcal{Y} \to \mathbb{R}_+$ is said to be ℓ-calibrated if there exists a link function $\psi: \mathbb{R}^d \to \mathcal{R}$ such that

$$\forall p \in \Delta_{\mathcal{Y}}, \quad \underbrace{\inf_{u \in \mathbb{R}^d : \psi(u) \notin \arg\min_t \langle p, \ell(t) \rangle}}_{\text{Inf over reports not linked to the argmin of the discrete loss.}} \langle p, L(u) \rangle > \inf_{u \in \mathbb{R}^d} \langle p, L(u) \rangle \ .$$

Consistency

Let $f_m: \mathcal{X} \to \mathbb{R}^d$ be the hypothesis learned from training on m samples drawn i.i.d. over P.

$er_P^L(f)$ is the expected loss L by predicting $f(X)$ when $(X, Y) \sim P$.

f_m is said to be L-consistent if

$$er_P^L(f_m) \to \inf_{f: \mathcal{X} \to \mathcal{R}} er_P^L(f) := er_P^{L,*}.$$

Losses are calibrated. Hypotheses are consistent.

Relating calibration and consistency

Let $\ell: \mathcal{R} \times \mathcal{Y} \to \mathbb{R}_+$ be a discrete loss.

A surrogate loss function $L: \mathbb{R}^d \times \mathcal{Y} \to \mathbb{R}_+$ is ℓ-calibrated

if and only if

there exists a link function $\psi: \mathbb{R}^d \to \mathcal{R}$ such that for all distributions P on $\mathcal{X} \times \mathcal{Y}$ and all sequences of surrogate hypotheses $f_m: \mathcal{X} \to \mathbb{R}^d$, we have

$$\underbrace{er_P^L[f_m] \to er_P^{L,*}}_{\text{Converging to optimal hypothesis for surrogate}} \Rightarrow \underbrace{er_P^\ell[\psi \circ f_m] \to er_P^{\ell,*}}_{\text{Linked hypothesis converges to optimal loss for discrete}}$$

Ramaswamy et al. (2015) Theorem 3, originally Tewari and Bartlett (2007) Theorem 2.

Pause

Questions so far?

Formalize properties.

- We use these to study calibration.

Properties

A property $\Gamma: \Delta_{\mathcal{Y}} \to \mathcal{R}$ is a function mapping probability distributions to reports.

If it's easier, think of $p \in \Delta_{\mathcal{Y}}$ as conditional probability.

A property $\Gamma: \Delta_{\mathcal{Y}} \to \mathcal{R}$ is elicitable if there is a loss function $L: \mathcal{R} \times \mathcal{Y} \to \mathbb{R}_+$ such that, for all $p \in \Delta_{\mathcal{Y}}$,

$$\Gamma(p) = \operatorname{argmin}_r \mathbb{E}_{Y \sim p} L(r, Y).$$

Here, we say the loss L elicits Γ.

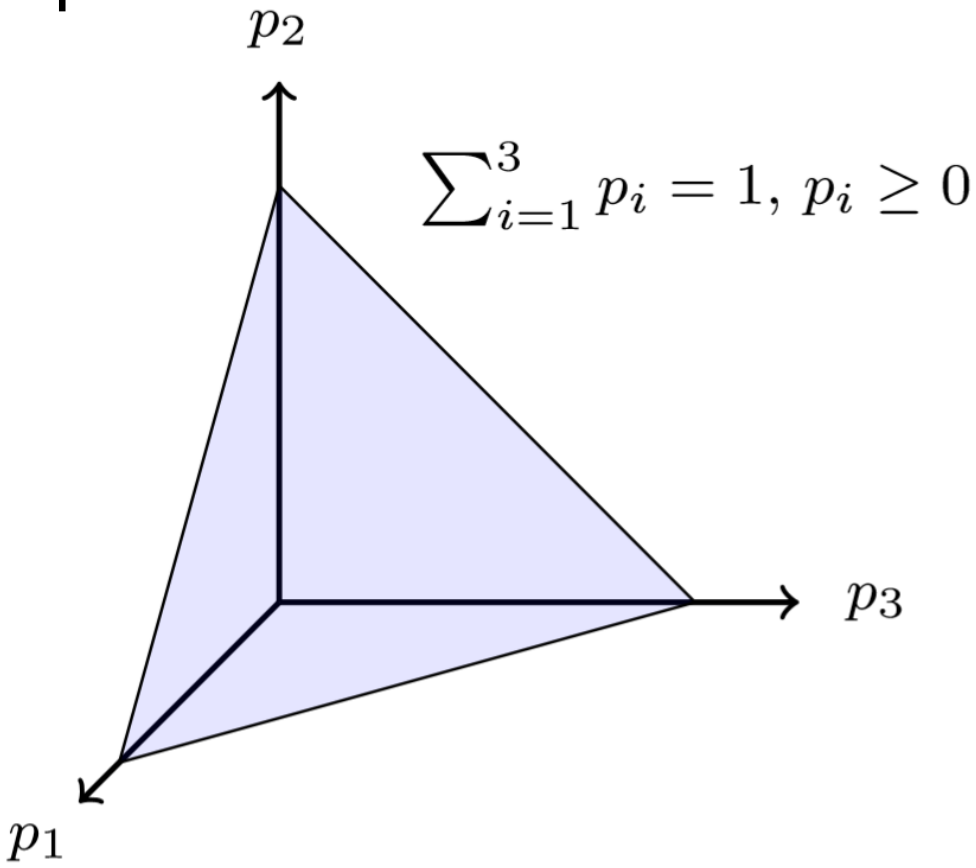
Elicitable properties have convex level sets. (Lambert and Shoham 2009)

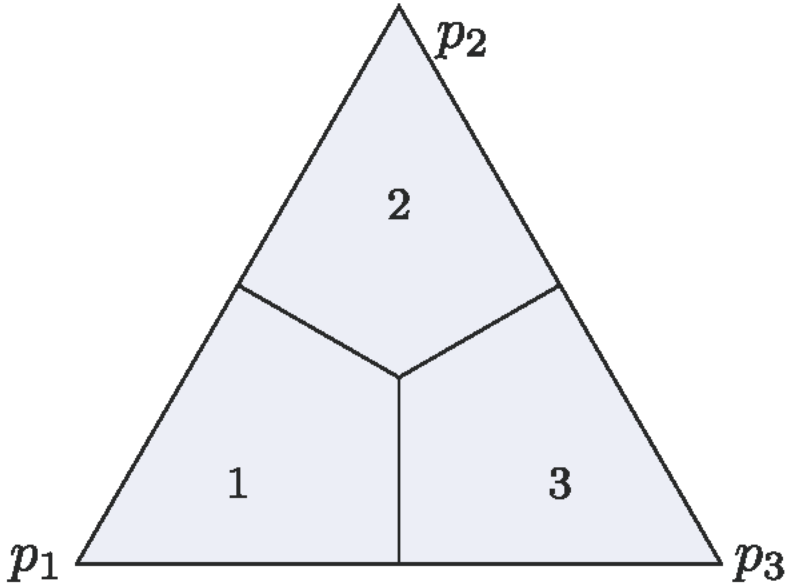
$$\Gamma_r = \{p \in \Delta_{\mathcal{Y}} : r \in \Gamma(p)\}$$

“Drawing” a property

n-simplex in $(n-1)$-dimensional space.

Example: n = 3





Calibration… in terms of properties

A property $\Gamma: \Delta_{\mathcal{Y}} \to \mathbb{R}^d$ and link $\psi: \mathbb{R}^d \to \mathcal{R}$ are ℓ-calibrated if

$$\{u_m\} \to \Gamma(p) \Rightarrow \mathbb{E}_p \ell(\psi(u_m), Y) \to \min_r \mathbb{E}_p \ell(r, Y).$$

i.e. The property value can always be linked to the argmin of loss.

Tool to study geometric properties of losses eliciting Γ.

Definition courtesy of Agarwal and Agarwal (2015)

Property Papers

Lambert, Shoham (2009): *Eliciting Truthful Answers to Multiple-choice questions.*

Finite properties are elicitable iff their level sets form a power diagram.

Agarwal, Agarwal (2015): *On Consistent Surrogate Risk Minimization and Property Elicitation.*

There's a connection between properties and surrogate losses.

Frongillo, Kash (2015): *On Elicitation Complexity.*

Every property is elicitable, but the question is *how* elicitable.

Calibrated surrogates

Positive Normal Sets.

Necessary Conditions.

Sufficient Conditions.

Relationship between positive normal sets and level sets of property.

Positive Normal Sets

Finite outcome setting: rewrite $L: \mathcal{R} \to \mathbb{R}^n$ be the vector of loss values should each outcome occur.

Linearity of expectation: rewrite $\mathbb{E}_p L(r, Y) = < p, L(r) >$.

Positive normal set

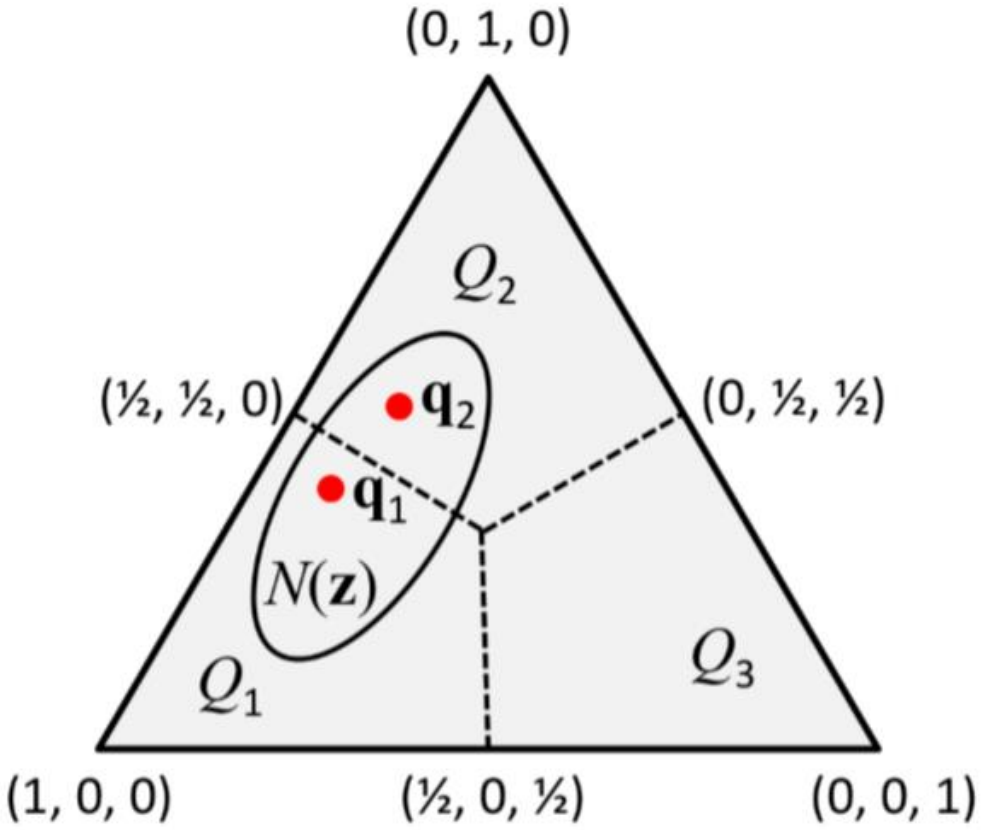
$$\mathcal{N}^L(\underbrace{u}_{\text{Outcome vector}}) := \left\{ \underbrace{p \in \Delta_{\mathcal{Y}}}_{\text{Distributions where}} : \underbrace{\langle p, u \rangle}_{\text{Expected loss on outcome vector}} = \underbrace{\inf_{u' \in \mathcal{S}^L} \langle p, u' \rangle}_{\text{Inf expected loss possible}} \right\}$$

Definition for sequences: sequence converges to inf.

Necessary Condition

Let ℓ be a discrete loss and let L be ℓ-calibrated. Let γ be the property elicited by ℓ.

For all $\mathrm{u} \in \mathcal{S}^L = conv(im(L))$, there exists an $r \in \mathcal{R}$ such that $\mathcal{N}^L(u) \subseteq \gamma_r$



Ramaswamy et al. (2015) Theorem 6

Sufficient Condition

Suppose there exists some finite set of $\{u_i\} \in \mathcal{S}^L$ such that $\cup_i \mathcal{N}^L(u_i) = \Delta_y$, and for each i there exists an $\mathrm{r}_j \in \mathcal{R}$ such that $\mathcal{N}^L(u_i) \subseteq \gamma_{r_j}$.

Then L is ℓ -calibrated.

Example: 0-1 loss and hinge

$$\mathcal{N}^{hinge}(\{2,0\}) = \left\{p \in \Delta_2 : p_1 \geq \frac{1}{2}\right\} \subseteq \gamma_1$$

$$\mathcal{N}^{hinge}(\{0,2\}) = \left\{p \in \Delta_2 : p_1 \leq \frac{1}{2}\right\} \subseteq \gamma_{-1}$$

$\mathcal{N}^{hinge}(\{0,2\})$	$\mathcal{N}^{hinge}(\{2,0\})$
γ_{-1}	γ_1

$p_1 = 0$ $\quad$ $p_1 = \frac{1}{2}$ $\quad$ $p_1 = 1$

Ramaswamy et al. (2015) Theorem 8

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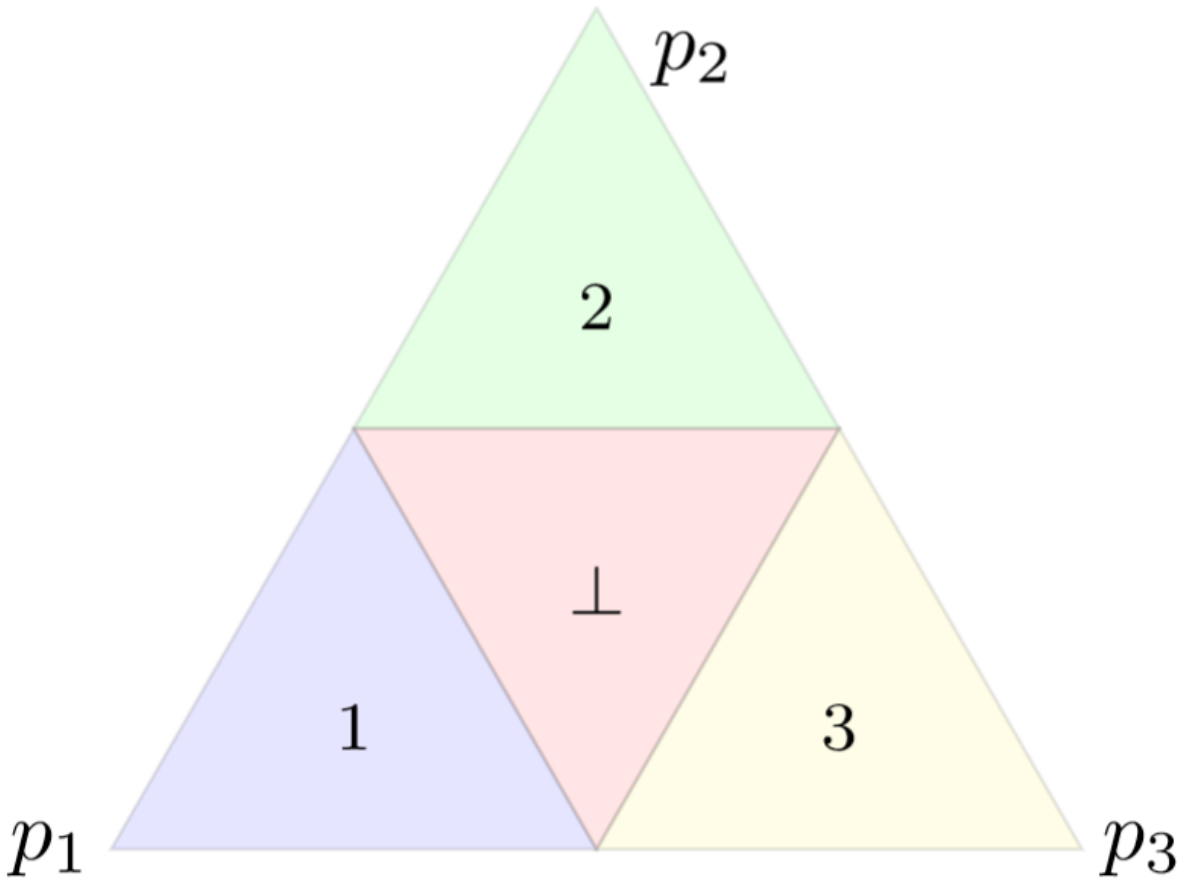
Case Study: Abstain

Situations where the cost of misclassification is high.

- College admissions.
- Medical diagnoses.

Discrete loss for this problem:

$$\ell^{\frac{1}{2}}(r, y) = \begin{cases} 0, & r = y \\ \frac{1}{2}, & r = \perp \\ 1, & r \neq y \text{ or } \perp \end{cases}$$



Historical calibrated surrogates

- Crammer Singer (2001)

$$L^{CS}(u,y) = \left(1 + \max_{j \neq y} u_j - u_y\right)_+$$

$$\psi^{CS}(u) = \begin{cases} argmax_{1 \leq i \leq n} u_i, & u_{[1]} - u_{[2]} > \tau \\ \perp, & otherwise \end{cases}$$

- One vs All (Rafkin, Klateau 2004)

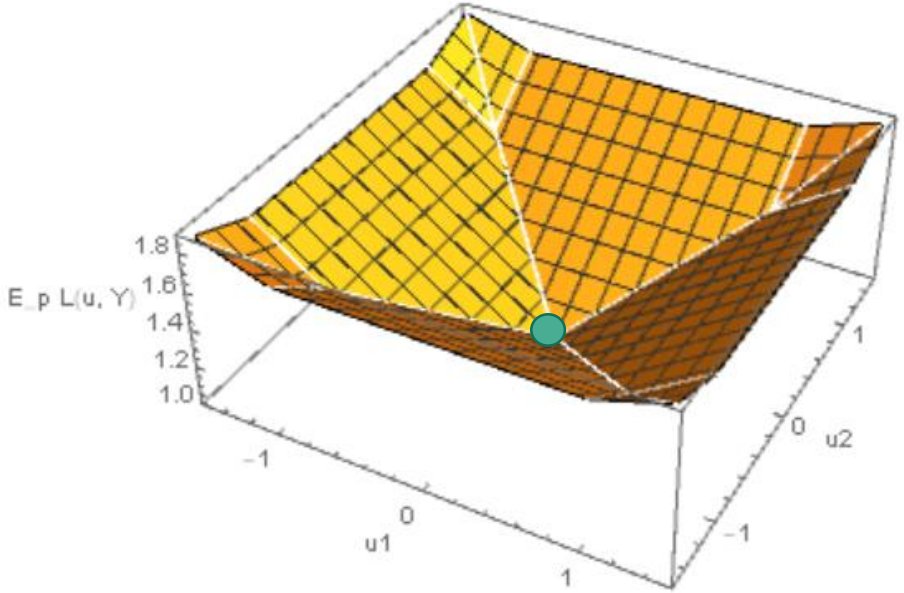
$$L^{OvA}(u,y) = \sum \mathbb{I}\{y = i\}(1 - u_i)_+ + \mathbb{I}\{y \neq i\}(1 + u_i)_+$$

$$\psi^{OvA}(u) = \begin{cases} argmax_{1 \leq i \leq n} u_i, & \max_j u_j > \tau \\ \perp, & otherwise \end{cases}$$

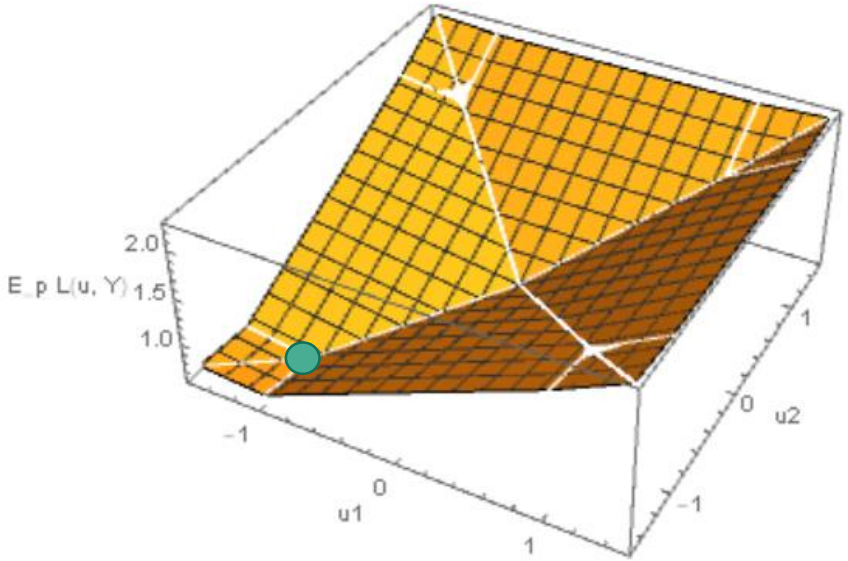
BEP Surrogate and Link

$$L^{1/2}(u, y) = \max_{j \in [d]} (B_j(y) u_j + 1)_+$$

$$\psi^{1/2} = \begin{cases} \perp & \min_{1 \le i \le d} |u_i| \le \tau \\ B^{-1}(\mathrm{sgn}(-u)) & \text{otherwise} \end{cases}$$



$p = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$



$p = (\frac{7}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10})$

Ramaswamy et al. (2018) Section 4

Why BEP?

BEP, CS, and OvA are all $\ell^{\frac{1}{2}}$-calibrated for abstain loss.

Convex surrogates takes report $u \in \mathbb{R}^d$.

BEP: $d = \log_2(n)$.
CS and OvA: $d = n$.

Why does dimension d matter?

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Dimensionality

Want algorithms that are efficient and accurate.

- Reducing dimension makes optimization problem more efficient.
- Calibration guarantees accuracy.

Elicitation Complexity

Elic(Γ) = Minimum dimension d such that Γ is elicitable by a d-dimensional loss.

Maybe Γ isn't itself 1-elicitable, but can be calculated by $\mathrm{g} \circ \hat{\Gamma}$, where $\hat{\Gamma}$ is 1-elicitable.

Then Elic(Γ) = 1.

Called indirect elicitation.

Example: $\Gamma(p) = \mathbb{E}_p[Y]^2$.

Elicit $\mathbb{E}_p[Y]$ and take $g: x \mapsto x^2$.

Convex Calibration Dimension

Special case of elicitation complexity.

ccdim(ℓ) = minimum dimension d such that there is a *convex ℓ-calibrated* surrogate $L: \mathbb{R}^d \times \mathcal{Y} \to \mathbb{R}$.

Ex: ccdim($\ell^{\frac{1}{2}}$) $\leq \log_2(n)$ because of BEP surrogate.

From Ramaswamy et al. (2015) Definition 10.

Bounds on CC dimension

Understood through Feasible Subspace Dimension.

Tight bound for properties where all the level sets intersect on interior of simplex.

$$\text{ccdim}(\ell) = n-1$$

Does not apply to abstain.

These results from Ramaswamy et al. (2015).

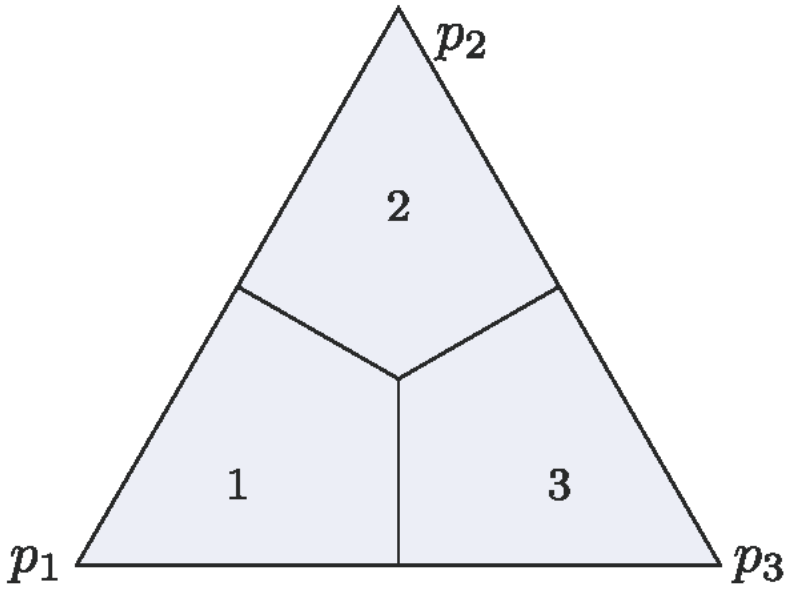
Feasible Subspace Dimension

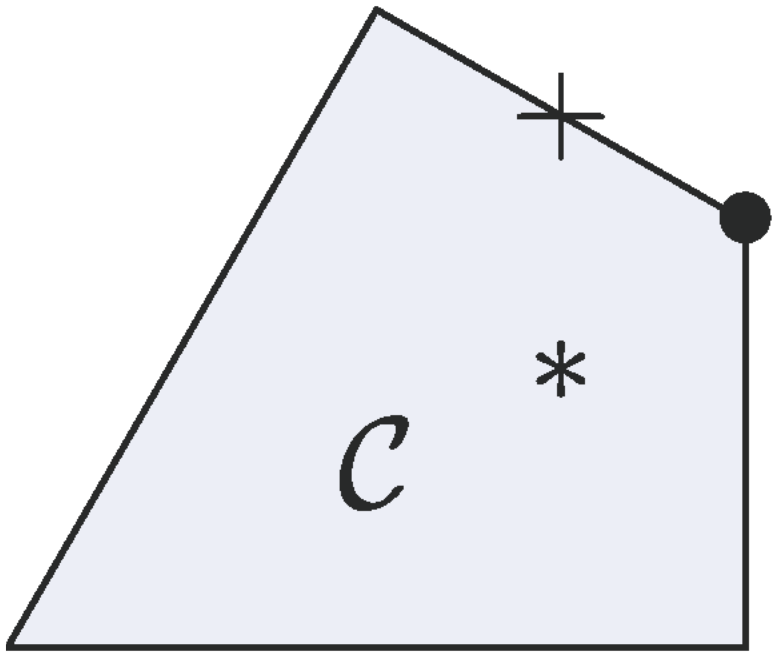
The feasible subspace dimension $\mu_{\mathcal{C}}(p)$ of a convex set $\mathcal{C}$ at the point $p \in \mathcal{C}$ is the dimension of $\mathcal{F}_{\mathcal{C}}(p) \cap -\mathcal{F}_{\mathcal{C}}(p)$.

- $\mathcal{F}_{\mathcal{C}}(p)$ is the cone of feasible directions of $\mathcal{C}$ at p.

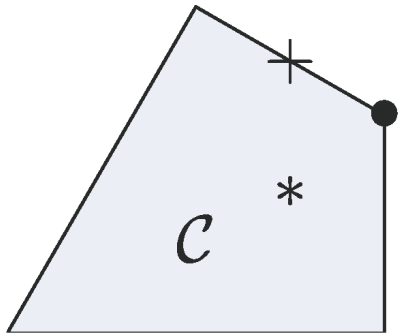
Essentially: the dimension of the smallest face of $\mathcal{C}$ containing p.

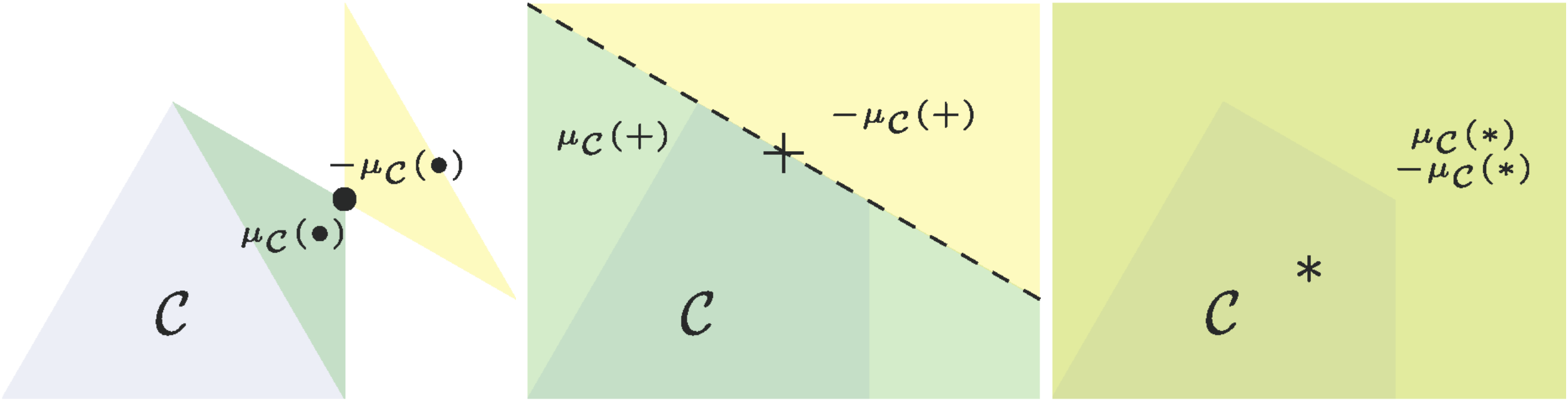
Feasible Subspace Dimension





Feasible Subspace Dimension





Lower bound on CC Dimension

Let ℓ be the discrete loss eliciting the finite property $\gamma: \Delta_{\mathcal{Y}} \to \mathcal{R}$.

For $p \in \Delta_{\mathcal{Y}}$ and r such that $p \in \gamma_r$, ccdim(ℓ) $\geq ||p||_0 - \mu_{\gamma_r}(p) - 1$.

Ramaswamy et al. (2016) Theorem 16

Skip proof sketch

Proof Sketch

Consider $p \in relint(\Delta_{\mathcal{Y}})$. Other case is a reduction to this.

Suppose surrogate $L: \mathbb{R}^d \to \mathbb{R}^n_+$ is ℓ-calibrated.

Want to show $d \geq ||p||_0 - \mu_{\gamma_r}(p) - 1$.

Want to show there is $\mathcal{H} \subseteq \Delta_{\mathcal{Y}}$ and $r' \in \mathcal{R}$ so that three things are true:

1. $p \in \mathcal{H}$
2. $\mu_{\mathcal{H}}(p) = n - d - 1$
3. $\mathcal{H} \subseteq \gamma_{r'}$

$p \in \gamma_{r'}$

$\mu_{\gamma_r}(p) = \mu_{\gamma_{r'}}(p) \geq \mu_{\mathcal{H}(p)} = n - d - 1$

Proof Sketch: Conditions 1 and 2

Construct $\mathcal{H}$ and r'.

Consider $u \in \mathcal{C}$ so that $\inf_{z \in \mathcal{S}^L} < p, z > = \inf_{u \in \mathcal{C}} < p, L(u) >$.

Claim: $\exists w_y \in \partial L_y(u)$ for all $y \in \mathcal{Y}$ so that $\sum_y p_y \overrightarrow{w_y} = \vec{0}$.

Set $A = [w_1, w_2, \ldots, w_n] \in \mathbb{R}^{d\ x\ n}$.

$$\mathcal{H} = \{q \in \Delta_{\mathcal{Y}} : Aq = \vec{0}\}$$

$\mu_{\mathcal{H}(p)} = nullity\left(\begin{bmatrix} A \\ \mathbb{1} \end{bmatrix}\right) \geq n - d - 1$ by Ramaswamy Lemma 15.

1. and 2. satisfied.

Condition 3 Intuition

We construct $\mathcal{H}$ using sequences that converge to u.

ϵ-subdifferential used to limit to the infimum.

They show $q \in \mathcal{H} \Rightarrow q \in \mathcal{N}^L(u)$ with a bunch of arithmetic.

Result follows since $\mathcal{N}^L(u) \subseteq \gamma_{r'}$ for some $r' \in \mathcal{R}$.

"Let ℓ be the discrete loss eliciting the finite property $\gamma: \Delta_{\mathcal{Y}} \to \mathcal{R}$. For $p \in \Delta_{\mathcal{Y}}$ and r such that $p \in \gamma_r$, ccdim(ℓ) $\geq ||p||_0 - \mu_{\gamma_r}(p) - 1$."

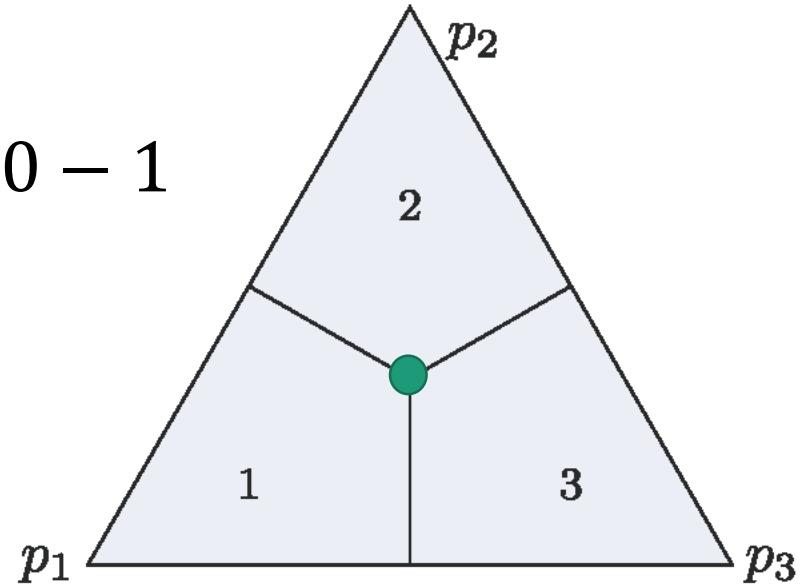
Mode: CC dimension tight bound

Trivial bound: $ccdim(0-1\ loss) \leq n-1$.

Previous result: For $p \in \gamma_r, ccdim(\ell) \geq \ ||p||_0 - \mu_{\gamma_r}(p) - 1$.

Take $p = \cap_r \gamma_r$.

$$ccdim(0-1\ loss) \geq n - 0 - 1$$

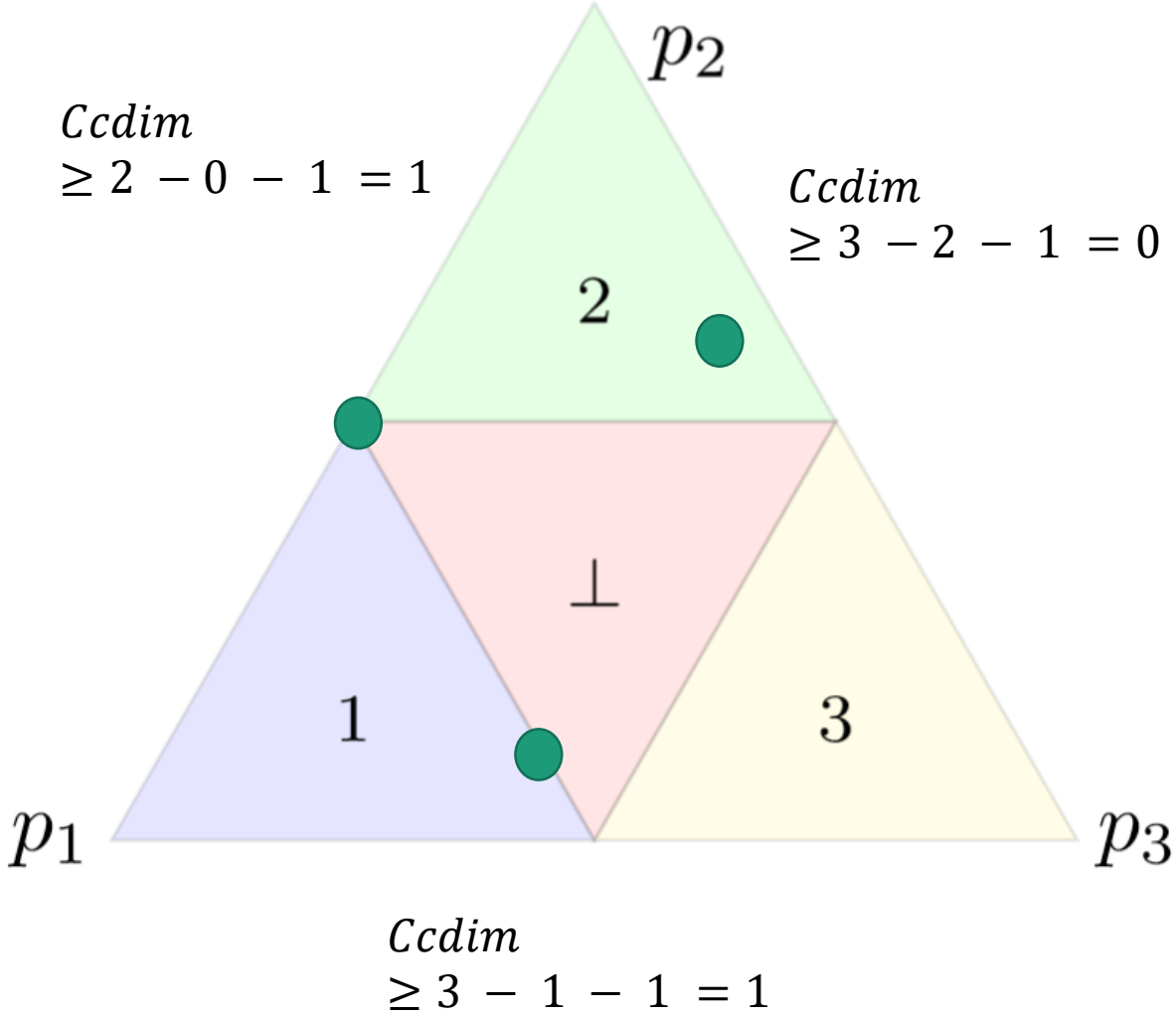


Thus, bound is tight.

Abstain: Unknown bounds

With abstain, $\cap_i \gamma_i = \emptyset$.

Thus, lower bound does not apply.



Calibrated surrogate papers

Ramaswamy, Agarwal (2015): *Convex calibration dimension for multiclass loss matrices.*

- Reducing surrogate dimension is important.
- Here are some bounds on dimension of calibrated convex surrogate losses.

Ramaswamy et al. (2018): *Consistent Algorithms for Multiclass classification with a reject option.*

- Abstain is cool, and here's a log(n) dimension consistent convex surrogate.

Other discrete examples

Ramaswamy et al. (2015): *Convex calibrated surrogates for hierarchical classification.*

Here's a calibrated surrogate for this problem using abstain surrogate.

Lapin et al. (2015): *Analysis and Optimization of Loss Functions for Multiclass, Top-k, and Multilabel Classification.*

Here are some surrogates for these problems. Not sure if they're calibrated.

Yu, Blaschko (2015): *The Lovász Hinge: A Novel Convex Surrogate for Submodular Losses.*

Here's how you go from submodular functions to convex surrogates.

Future Work

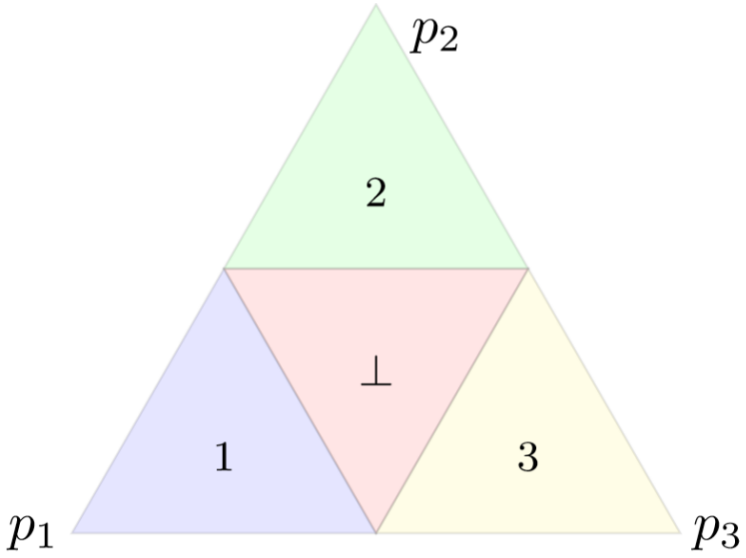
Embedding Dimension.

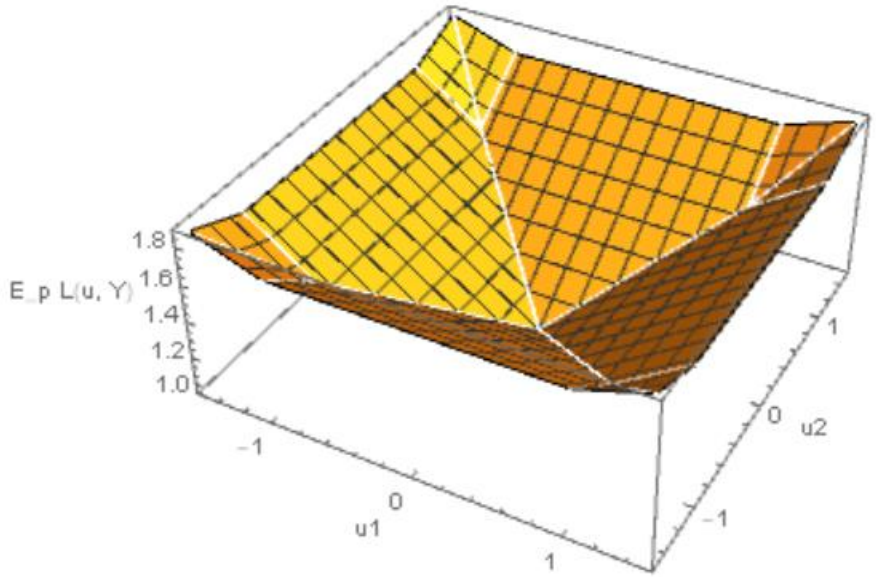
Constructing Link Functions.

Approximate Elicitation.

Summary

$$\ell^{\frac{1}{2}}(r,y) = \begin{cases} 0, & r = y \\ \frac{1}{2}, & r = \perp \\ 1, & r \neq y \text{ or } \perp \end{cases}$$





Thank you, questions?